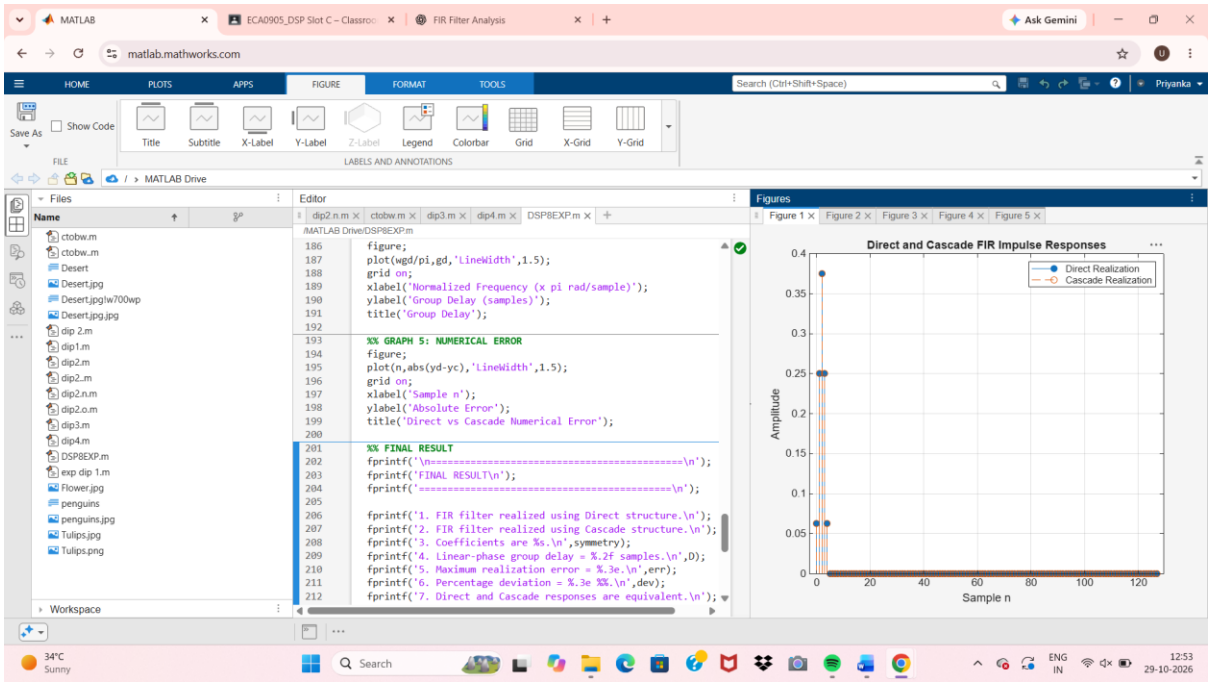
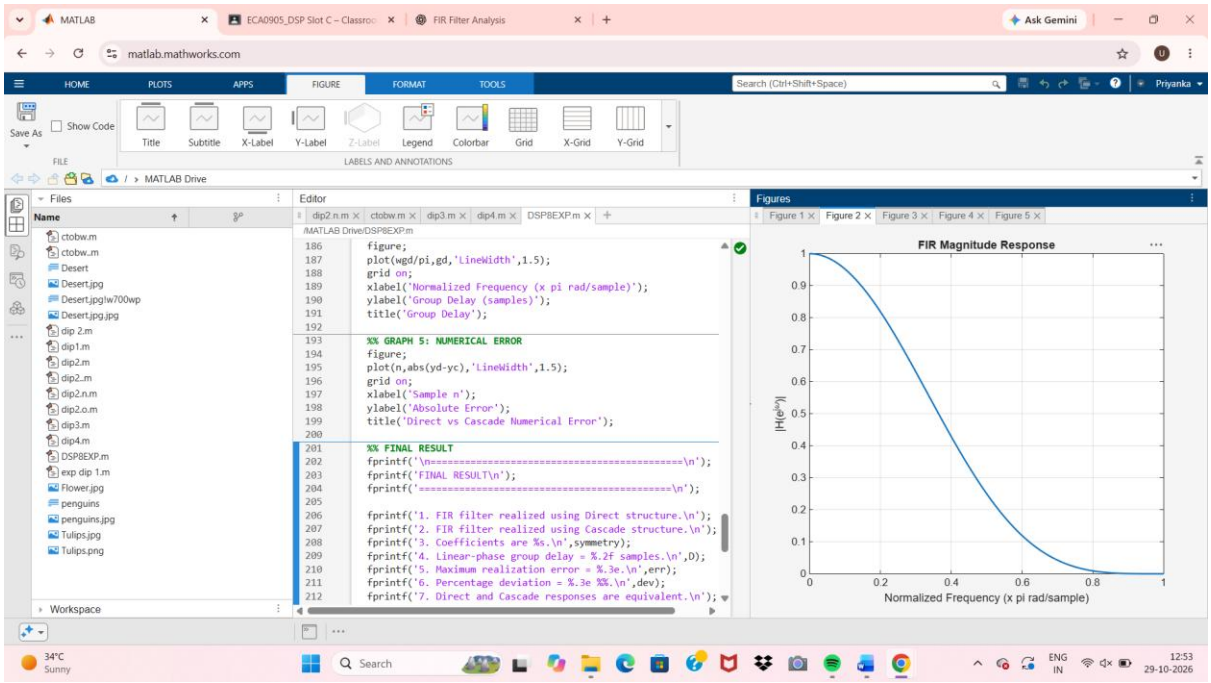
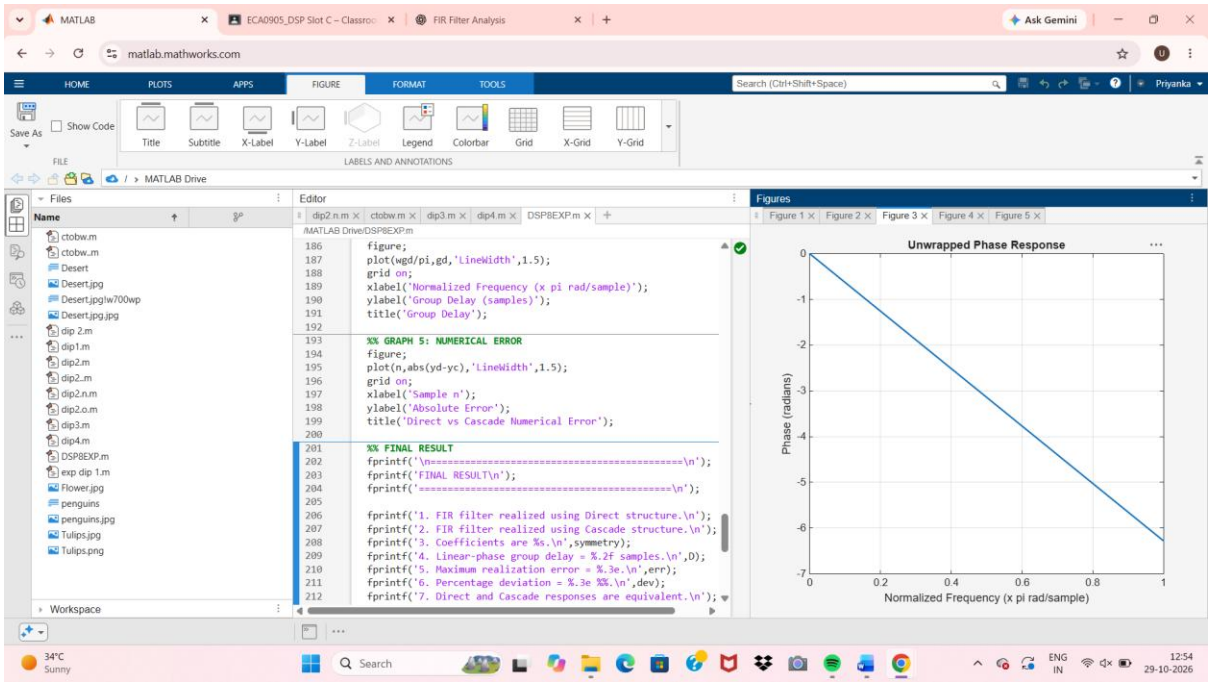
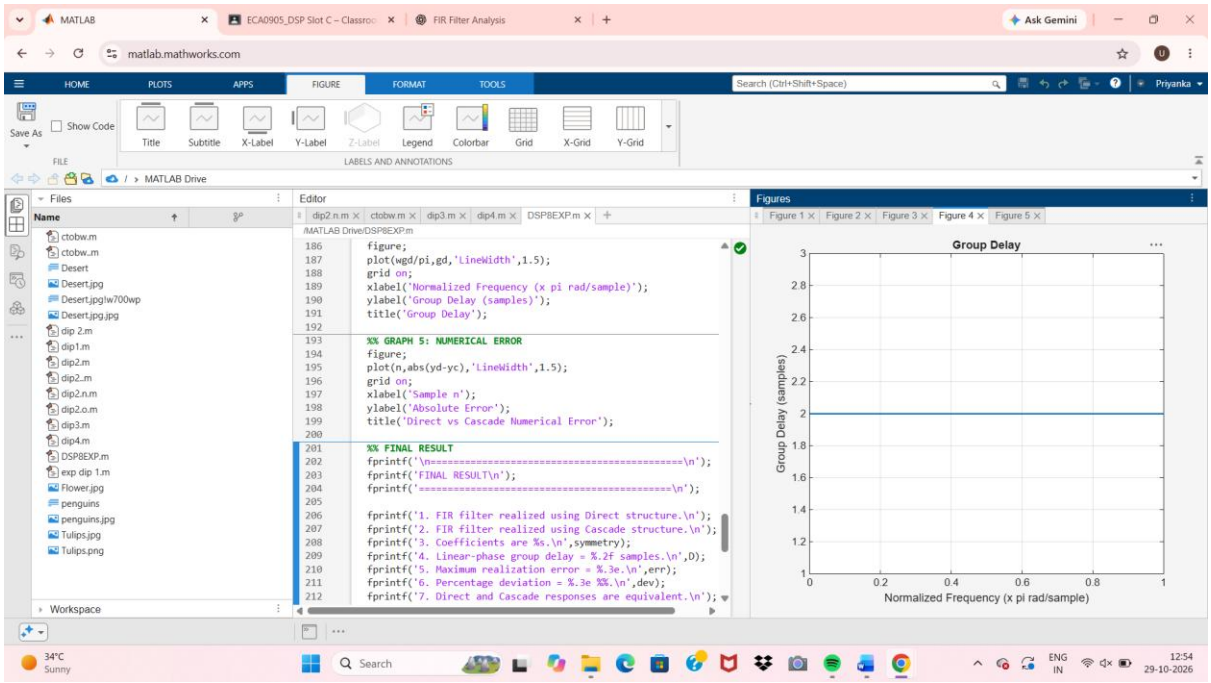
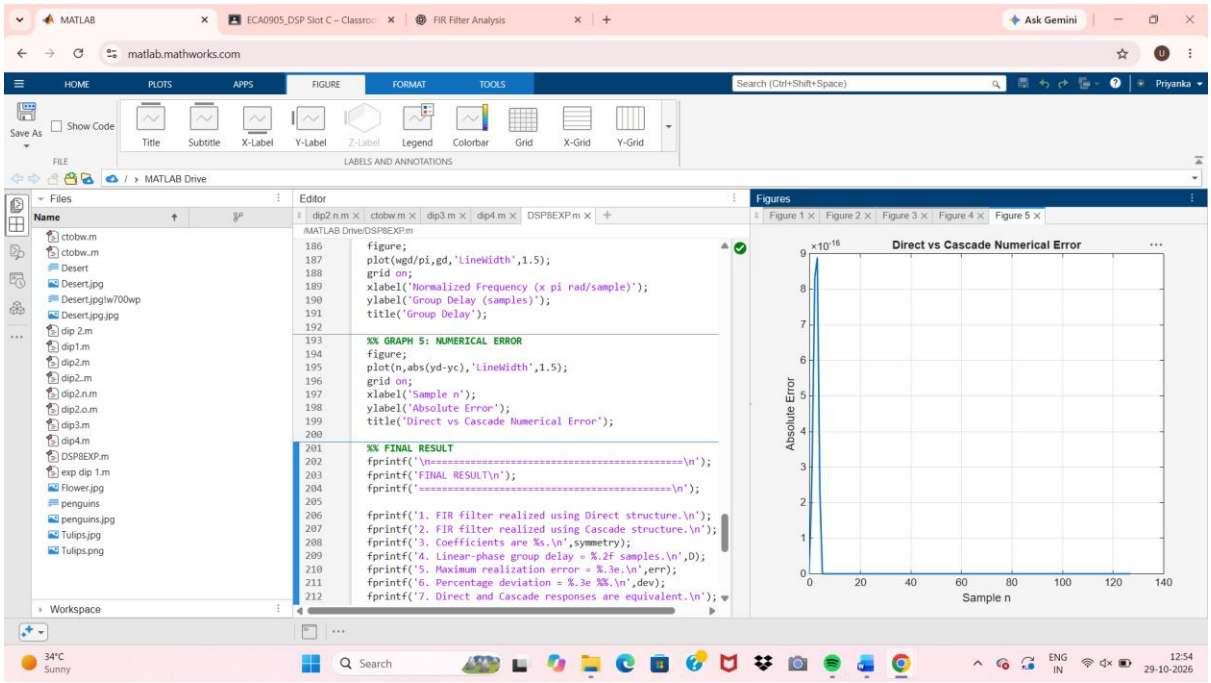

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figure;
plot(wgd/pi,gd,'LineWidth',1.5);
grid on;
xlabel('Normalized Frequency (x pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
%% GRAPH 5: NUMERICAL ERROR
figure;
plot(n,abs(yd-yc),'LineWidth',1.5);
grid on;
xlabel('Sample n');
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title('Direct vs Cascade Numerical Error');
%% FINAL RESULT
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fprintf('3. Coefficients are %s.\n',symmetry);
fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
fprintf('5. Maximum realization error = %.3e.\n',err);
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
fprintf('7. Direct and Cascade responses are equivalent.\n');
Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Direct and Cascade FIR Impulse Responses
Direct Realization
Cascade Realization
Amplitude
Sample n
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Figures
Figure 1
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FIR Magnitude Response
|H(e^jω)|
Normalized Frequency (x pi rad/sample)
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Figures
Figure 1
Figure 2
Figure 3
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Figure 5
Unwrapped Phase Response
Phase (radians)
Normalized Frequency (x pi rad/sample)
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Figure 5
Group Delay
Group Delay (samples)
Normalized Frequency (x pi rad/sample)
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Figure 3
Figure 4
Figure 5
Direct vs Cascade Numerical Error
×10^-16
Absolute Error
Sample n
34°C
Sunny
Search
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